BRIAN STEWART

MECHANICAL ENGINEER, EIT

stewartbd23@gmail.com

linkedin.com/in/brianstewart23

(720) 737 - 0019

SKILLS

CAD & MECHANICAL

- **SolidWorks (CSWA)**, **ANSYS** used on various projects involving **FEA**, **SketchUp**
- SolidWorks 2D engineering drawings
- **XFLR5** used in **GN&C** stability projects

ELECTRICAL & SOFTWARE

- **C++, C, MATLAB, Julia, Arduino**
- **KiCAD** used in **PCB design**
- Soldering, Wiring, Electrical Testing

PROJECTS

ROCKET TRAILER

- Launches hybrid rocket while providing NO2 and electronics

ROBOT

- Collects ping pong balls and shoots them into nets

PIPSQUEAK ENGINE

- Converts compressed air into rotational motion

G-MOTOR ROCKET

- Launches 1200 ft into air with a solid motor

RC PLANE

- Features servo motors for controls and variable speed propeller

EDUCATION

Brigham Young University

- **Mechanical Engineering**
- **Aug 2018 - Apr 2025**
- **Engineer in Training (EIT) - Passed FE Exam on 10/29/2024**

EXPERIENCE

ROHM | Mechatronic Engineering Intern | Novi, MI

May - Aug 2024

- Developed automotive lighting and motor software using **C** programming language and **Arduino** IDE to reduce the complexity of vehicle software by 75%
- Created a mechatronic demo using **KiCAD** software and **PCB design** to represent ROHM at the Electronica Conference in Germany
- Tested electronic components using data acquisition tools such as oscilloscopes and multimeters to determine reliability

BYU IT Department | Technical Support Specialist | Provo, UT

Aug 2023 - Present

- **Troubleshoot** technical issues using windows diagnostics tools and active directory to provide support for over 200 employees
- Re-image, inventory and update over 300 Microsoft laptop and desktop equipment
- Contribute to **project management** of the team by using **Jira**, **Confluence**, and **Excel,** speeding up solution times

Solcius | Sales Commission Specialist | Provo, UT

Feb 2022 - Feb 2023

- Drafted over 200 CAD houses and solar layouts with **SketchUp,** giving customers and sale reps power readings with 90% accuracy
- Managed **quality control** through **Salesforce** to assist in shortening Texas Region solar processing time by 50%
- Systemized solar underwriting process and resolved over 100 issues reported by sales representatives over phone and email

BYU FLOW Lab | Undergraduate Research Assistant | Provo, UT

Jun 2021 - Jan 2022

- Developed code using **Julia** programming language to optimize the layout of a wind farm given a specified area for maximum power production
- Produced a research paper on the analysis of wind turbine wake interaction using **mathematical modeling** to prove ideal scenarios for optimal wind farm layout
- Produced a research paper on the “Leapfrogging Vortices” effect using **engineering analysis** to simulate the effect with 0.00001 second resolution

The Church of Jesus Christ | Full-Time Volunteer | Belgium/Netherlands

May 2019 - Mar 2020

- Led over 10 weekly meetings for a team of eight volunteers in eastern Netherlands district
- Served over 100 people at two different food banks on a weekly basis to meet community needs
- Taught English classes to immigrants from 4 different countries through weekly meetings allowing them to better assimilate to European culture

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MECHANICAL ENGINEER

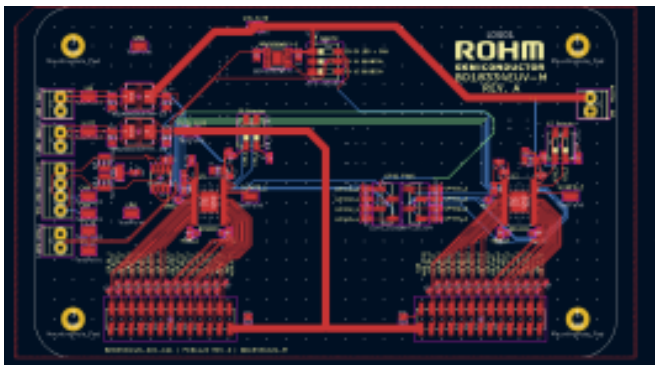
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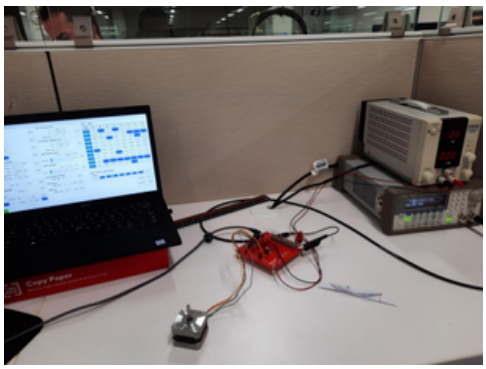
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AUTOMOTIVE MECHATRONICS DEMO - ROHM







What?

- Create and test a working LED driver demo that controls automotive lights and motors from one MCU
- Reduces electrical complexity by 75%

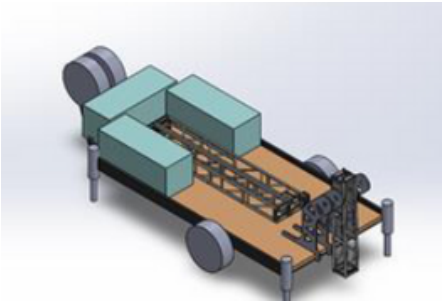
How?

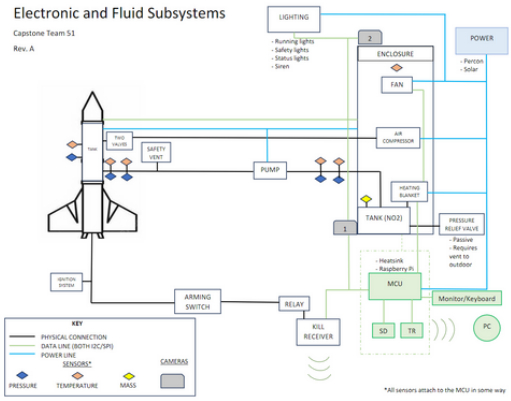
- Utilized **KiCAD** software to model and analyze **PCB design** and circuits
- Wrote code in **C language** on **Arduino** IDE to control LEDs and motor

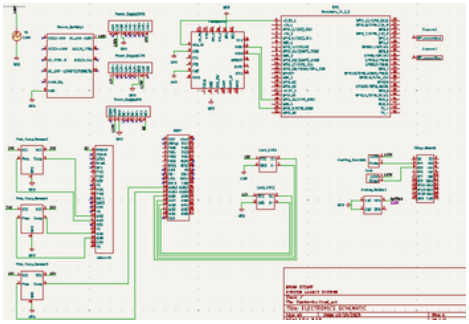
Results

- The demo functioned as designed and reduced electrical complexity
- Presented at Electronica conference in Germany in November

HYBRID ROCKET TRAILER - BYU ROCKETRY CLUB







What?

- Design and build a functional hybrid rocket trailer that transports NO2 and can successfully launch a rocket off its back
- Conform to all safety standards and rules for Spaceport launch competition

How?

- Interacted with stakeholders to identify and document requirements
- Assisted in developing and maintaining operational procedures
- Created 3D models for electrical system using **Solidworks**
- Created schematics and high-level models using **KiCAD** and **Visio**

Results

- Currently building trailer with a team of 10 to be finished and functioning in February of 2025
- **Team leader** of the electronics subsystem, leading meetings and project progression
- Finalized critical design complete with CAD models and designs

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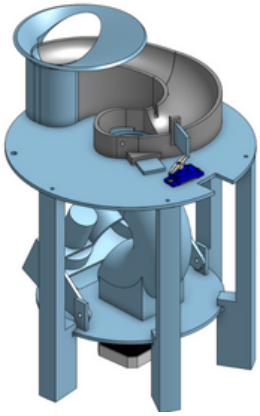
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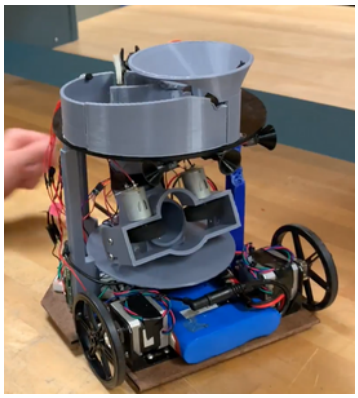
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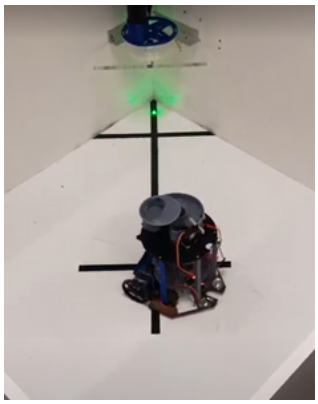
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PING PONG SHOOTING ROBOT - BYU







What?

- Build a robot that is able to acquire ping pong balls, store them, read IR sensors and shoot different colored balls into corresponding nets

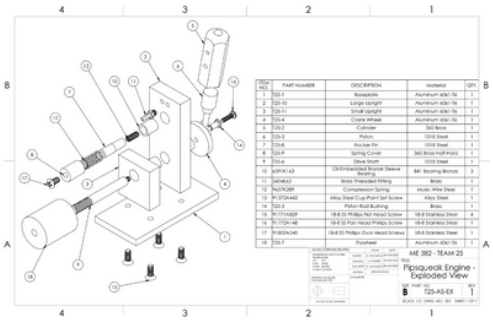
How?

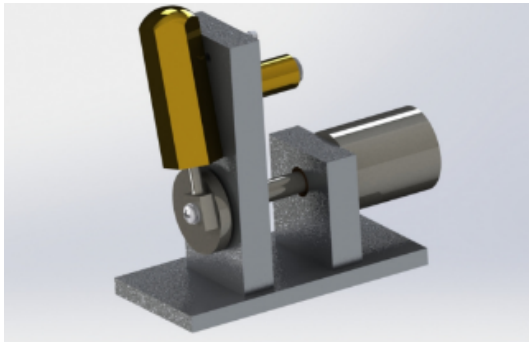
- Helped create 3D models in **SolidWorks**
- Wrote code in **C language** to allow communication between robot and environment (**sensors**)
- Tested components, battery, stepper motors, and shooter

Results

- Robot shot balls with 85% accuracy and functioned according to design
- Shot 14 balls into the nets in allotted time of 1 minute and 30 seconds

PIPSQUEAK ENGINE - BYU







What?

- Design and build a single piston engine attached to a flywheel that is powered by compressed air.
- Run continuously while increasing air pressure until failure

How?

- Helped create 3D models and drawings in **SolidWorks**
- Manufactured parts using **lathing**, **milling**, and **threading**
- Helped create Bill of Materials and document manufacturing processes

Results

- Reached a high of 30 psi which was higher than expected
- Functioned as intended and delivered rotational energy from compressed air

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RC AIRPLANE



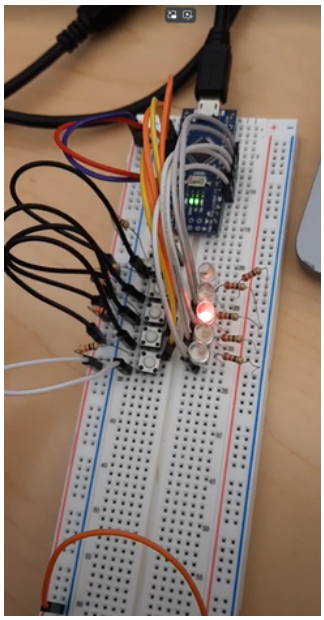


What?

- Created a remote-control airplane from scratch with propeller, motor with electronic speed control, battery, and servo motors for aileron, elevator and rudder control
- Plane was made out of foam core and dowel rods for extra support. Landing gear was also included

ROBOTIC VEHICLE - BYU





What?

- Design and build a robot that can stops when abject is in front of the vehicle
- Can detect light and drive towards light using IR sensor

How?

- Utilized breadboard circuitry and Arduino nano microcontroller to integrate environment with robotic action
- Coded in Arduino IDE using C programming language

G-MOTOR ROCKETRY COMPETITION





What?

- Launch a solid motor rocket 1,000 ft in the air and safely parachute down

Results

- Launched 1,200 ft in air, 200 feet higher than expected